

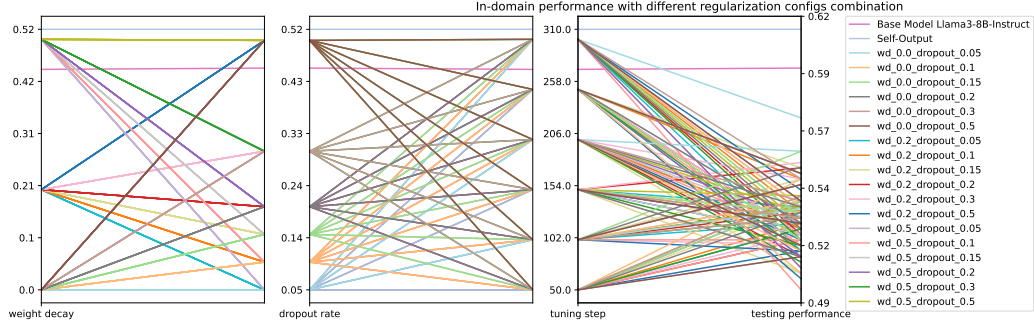


Figure 11: The target task performance of all combinations of regularization parameters on MBPP ground-truth data model (based on Llama 3 8B Instruct) and comparison between the original Llama 3 8B Instruct model's performance on MBPP testing dataset.

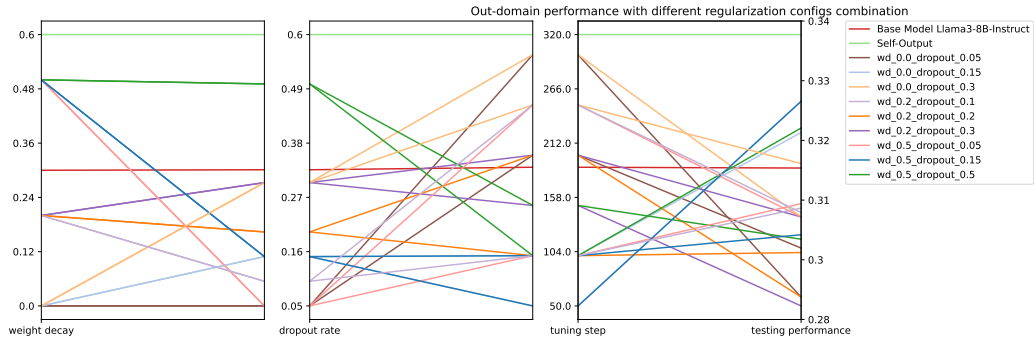


Figure 12: The non-target MATH task performance of all combinations of regularization parameters on MBPP ground-truth data model (based on Llama 3 8B Instruct) and comparison between the original Llama 3 8B Instruct model's performance on MATH testing dataset.

shows, we found that some of the results perform better than the original Llama 3 8B Instruct in non-target task, but their target task performances are found worse than the base model's performance. It implies that the widely adopted regularization techniques are hard to optimized for both target task and non-target task performance.

K Rank analysis on LoRA weights

To further understand this behavior, we examine the relationship between weight updates and the information-theoretic properties of the training data. For a given sequence of tokens, we define the token perplexity as $I(x, y) = -\sum_{t=1}^T \log P(y_t | x_{<t}; \theta)$, where $P(y_t | x_{<t}; \theta)$ represents the model’s predicted probability for the true token y_t given the preceding context. In LoRA, weight updates for each transformer layer are parameterized as low-rank matrices $\Delta W = BA$, where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $r \ll \min(d, k)$. Through Singular Value Decomposition (SVD), $\Delta W = U\Sigma V^\top$, we can analyze both the magnitude of singular values in Σ and its effective rank given a threshold τ .

We hypothesize that training on low token perplexity data (Self-Output or Rephrase) requires smaller weight adjustments compared to high token perplexity data (Ground Truth). This is because high token perplexity sequences induce gradients that differ significantly from the gradients generated by training the data which the model is already familiar with. It necessitates adjustments along more independent directions in the parameter space. Our empirical analysis, comparing LoRA weight updates across training conditions with identical hyperparameters and training steps, confirms this hypothesis: Ground Truth data consistently produces updates with higher effective rank, as evidenced by the singular value spectrum of ΔW . This aligns with our theoretical framework, where high perplexity tokens require the model to adjust parameters along more dimensions to minimize the loss for these unexpected sequences. In Figure 13 and Figure 14, we identify effective singular value ranks on layers 17th to 32nd on self-attention output projection of Llama 3 8B Instruct LoRA weights and find that Ground Truth fine-tuning enables more ranks to learn with higher values for each attention, indicating more perturbations on training to degrade the general capabilities to fit a target task.

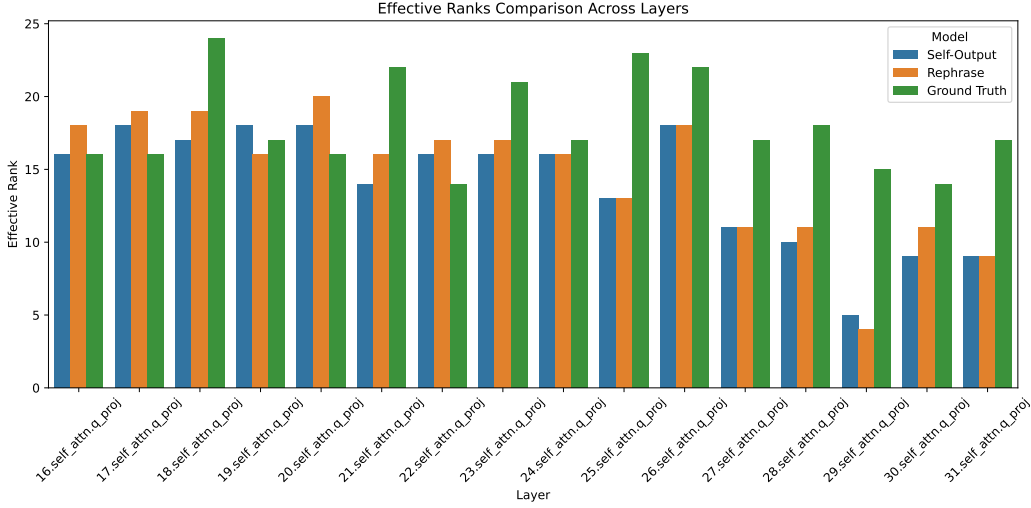


Figure 13: Effective singular value rank on layers 17th to 32nd on self-attention output projection Llama 3 8B Instruct LoRA weights finetuned on MBPP at step 164

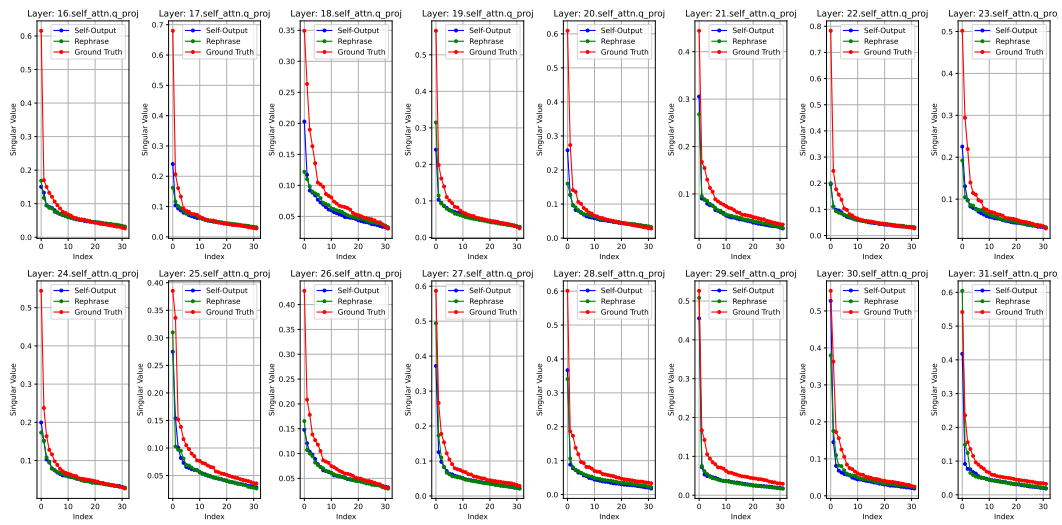


Figure 14: Singular values on layers 17th to 32nd on self-attention output projection LoRA weights finetuned on MBPP at step 164